

Second Partial

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Chain rule

Definition of the chain rule

The chain rule allow us to compute the derivative of a composition of functions $f(x) = (h \circ g)(x) = h(g(x))$. The rule goes as follows,

Leibniz notation

$$\frac{df(x)}{dx} = \left[\frac{dh(x)}{dx} \circ g(x) \right] \frac{dg(x)}{dx}$$

Lagrange notation

$$f'(x) = h'(g(x)) \cdot g'(x)$$

Chain rule

Examples

Let's compute the derivative of $f(x) = \ln(\cos(x))$.

$$\begin{aligned} f'(x) &= \ln'(\cos(x)) \cdot \cos'(x) \\ &= \frac{1}{\cos(x)} \cdot -\sin(x) \\ &= -\tan(x) \end{aligned}$$

Chain rule

Examples

Let's compute the derivative of $f(x) = \cos(2\pi x - \pi)$.

$$\begin{aligned} f'(x) &= \cos'(2\pi x - \pi) \cdot (2\pi x - \pi)' \\ &= -\sin(2\pi x - \pi) \cdot (2\pi) \\ &= -2\pi \sin(2\pi x - \pi) \end{aligned}$$

Chain rule

Examples

Let's compute the derivative of $f(x) = e^{-\frac{x^2}{4}}$.

$$\begin{aligned} f'(x) &= \left[\frac{de^x}{dx} \circ -\frac{x^2}{4} \right] \cdot \left(-\frac{x^2}{4} \right)' \\ &= e^{-\frac{x^2}{4}} \cdot \left(-\frac{x}{2} \right) \\ &= -\frac{x}{2} e^{-\frac{x^2}{4}} \end{aligned}$$

Chain rule

Class Activity

Compute the derivative of the following functions using the chain rule:

$$f(x) = \ln(e^x), \quad g(x) = e^{\cos(x)}$$

Rule of derivatives

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

Chain rule

$$\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$$

Quotient rule

$$\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$$

Product rule

$$\frac{d}{dx}u(x)v(x) = u'v + uv'$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Product rule

First we identify that is a product of a composition of functions. It is a product, because we can write $f(x) = u(x)v(x)$. Where $u(x) = e^{-x^2}$, and $v(x) = \sin(2\pi x)$.

Chain rule

Then, looking at $u(x)$ and $v(x)$ we can see that is a composition of functions, $u(x) = (e^x \circ -x^2)(x) = e^{-x^2}$ and $v(x) = (\sin(x) \circ 2\pi)(x) = \sin(2\pi x)$.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Product rule

Let's start with the product rule,

$$\begin{aligned} f'(x) &= u'v + uv' \\ &= \left(\frac{d}{dx} e^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \end{aligned}$$

Now, it is time to use the chain rule to compute the derivatives of u and v .

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Chain rule: e^{-x^2}

As we identify before, $e^{-x^2} = e^x \circ -x^2$, therefore,

$$\begin{aligned}\frac{d}{dx}e^{-x^2} &= \frac{d}{dx}e^x \Big|_{x=-x^2} \left(\frac{d}{dx} -x^2 \right) \\ &= e^x \Big|_{x=-x^2} (-2x) \\ &= e^{-x^2} (-2x) \\ &= -2xe^{-x^2}\end{aligned}$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Replace

Let's replace the previous result into the solution,

$$\begin{aligned} f'(x) &= \left(\frac{d}{dx} e^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \\ &= \left(-2xe^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \end{aligned}$$

Let's continue with $\frac{d}{dx} \sin(2\pi x)$.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Chain rule: $\sin(2\pi x)$

As we identify before, $\sin(2\pi x) = \sin(x) \circ 2\pi x$, therefore,

$$\begin{aligned}\frac{d}{dx} \sin(2\pi x) &= \frac{d}{dx} \sin(x) \Big|_{x=2\pi x} \left(\frac{d}{dx} 2\pi x \right) \\ &= \cos(x) \Big|_{x=2\pi x} \cdot 2\pi \\ &= \cos(2\pi x) \cdot 2\pi \\ &= 2\pi \cos(2\pi x)\end{aligned}$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Replace

Let's replace the previous result into the solution,

$$\begin{aligned} f'(x) &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x)\right) \\ &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} (2\pi \cos(2\pi x)) \end{aligned}$$

Now we can move on to simplify the expression.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Simplify

$$\begin{aligned} f'(x) &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} (2\pi \cos(2\pi x)) \\ &= 2e^{-x^2} [-x \sin(2\pi x) + \pi \cos(2\pi x)] \\ &= 2e^{-x^2} [\pi \cos(2\pi x) - x \sin(2\pi x)] \end{aligned}$$

$$f'(x) = 2e^{-x^2} [\pi \cos(2\pi x) - x \sin(2\pi x)]$$

Class Activity

Compute the derivative of the following functions using the rules.

$$f(x) = e^{-\frac{x^2}{4}} \sin\left(\frac{\pi}{2}x + \pi\right), \quad g(x) = \sin^2(x)$$

Answers

$$f'(x) = \frac{1}{2}e^{-x^2/4} \left[\pi \cos\left(\frac{\pi}{2}x + \pi\right) - x \sin\left(\frac{\pi}{2}x + \pi\right) \right]$$

$$g'(x) = 2 \sin(x) \cos(x)$$

Rule of derivatives

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

Quotient rule

$$\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$$

Chain rule

$$\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$$

Product rule

$$\frac{d}{dx}u(x)v(x) = u'v + uv'$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Quotient rule

The function $f(x)$ is a quotient of functions, where $u = x$ and $v = (x^2 + 1)^{3/2}$. By applying the quotient rule, $\frac{d}{dx}u/v = \frac{1}{v^2}(u'v - uv')$, we get,

$$\begin{aligned} f'(x) &= \frac{1}{\left[(x^2 + 1)^{3/2}\right]^2} \left[\left(\frac{d}{dx}x \right) (x^2 + 1)^{3/2} - x \left(\frac{d}{dx}(x^2 + 1)^{3/2} \right) \right] \\ &= \frac{1}{(x^2 + 1)^3} \left[\left(\frac{d}{dx}x \right) (x^2 + 1)^{3/2} - x \left(\frac{d}{dx}(x^2 + 1)^{3/2} \right) \right] \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

$$f'(x) = \frac{1}{(x^2+1)^3} \left[\left(\frac{d}{dx} x \right) (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right]$$

Power rule: $\frac{d}{dx} x$

$$\frac{d}{dx} x = 1x^{1-1} = 1x^0 = 1 \cdot 1 = 1$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Replace

$$\begin{aligned} f'(x) &= \frac{1}{(x^2+1)^3} \left[\left(\cancel{\frac{d}{dx} x}^1 \right) (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] \\ &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Chain rule: $\frac{d}{dx}(x^2+1)^{3/2}$

Here, $u = x^{3/2}$ and $v = x^2 + 1$, so that $(u \circ v)(x) = (x^2 + 1)^{3/2}$. With this, we can apply the chain rule, $\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$,

$$\begin{aligned}\frac{d}{dx}(x^2+1)^{3/2} &= \left. \frac{d}{dx}x^{3/2} \right|_{x=x^2+1} \cdot \left[\frac{d}{dx}(x^2+1) \right] \\ &= \left. \frac{3}{2}x^{1/2} \right|_{x=x^2+1} \cdot [2x+0]\end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Chain rule: $\frac{d}{dx}(x^2+1)^{3/2}$

$$\begin{aligned}\frac{d}{dx}(x^2+1)^{3/2} &= \frac{3}{2}x^{1/2} \Big|_{x=x^2+1} \cdot [2x+0] \\ &= \frac{3}{2}\sqrt{x^2+1} \cdot 2x \\ &= 3x\sqrt{x^2+1}\end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Replace

$$\begin{aligned} f'(x) &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] 3x\sqrt{x^2+1} \\ &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(3x\sqrt{x^2+1} \right) \right] \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Simplify

$$\begin{aligned} f'(x) &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(3x\sqrt{x^2+1} \right) \right] \\ &= \frac{1}{(x^2+1)^3} \left[(x^2+1)^{3/2} - 3x^2\sqrt{x^2+1} \right] \end{aligned}$$

$$f'(x) = \frac{1-2x^2}{\sqrt{(x^2+1)^5}}$$

Class Activity

Compute the derivative of the following functions:

$$f(x) = \frac{2}{1 + e^{-2(x-4)}}$$

$$f'(x) = \frac{4e^{-2(x-4)}}{[1 + e^{-2(x-4)}]^2}$$

$$g(x) = \frac{2x}{(x^2 + 4)^{3/2}}$$

$$g'(x) = \frac{8 - 4x^2}{\sqrt{(x^2 + 4)^5}}$$

Power rule: $\frac{d}{dx} x^n = nx^{n-1}$

Chain rule: $\frac{d}{dx} u(v(x)) = u'(v(x)) \cdot v'(x)$

Quotient rule: $\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$

Product rule: $\frac{d}{dx} u(x)v(x) = u'v + uv'$

Derivative of Exponential functions $f(x) = n^x$

We know how to compute the derivative of x^n using the power rule: $\frac{d}{dx}x^n = nx^{n-1}$. What if we change the symbols? $x^n \rightarrow n^x$. How we can compute the derivative of an exponential function?

Hint: Express the exponential function in terms of e^x

We know that $\frac{d}{dx}e^x = e^x$. Therefore, the big idea is to express the function n^x in terms of e^x .

Example: $f(x) = 2^x$

Express 2^x in terms of e^x

We can use the property of inverse functions: $e^{\ln(x)} = x$. But instead of using x , let's use 2^x ,

$$e^{\ln(2^x)} = 2^x.$$

Now, by using properties of logarithms, $\ln(x^n) = n \ln(x)$,

$$e^{x \ln(2)} = \left[e^{\ln(2)} \right]^x = 2^x.$$

Example: $f(x) = 2^x$

Use the chain rule to compute the derivative

Now, we can compute the derivative of $f(x)$

$$\begin{aligned}\frac{d}{dx} 2^x &= \frac{d}{dx} e^{x \ln(2)} \\ &= \left. \frac{d}{dx} e^x \right|_{x=x \ln(2)} \cdot \left[\frac{d}{dx} (x \ln(2)) \right] \\ &= e^{x \ln(2)} \cdot \left[\frac{d}{dx} (x \ln(2)) \right]\end{aligned}$$

Example: $f(x) = 2^x$

Use the product rule to compute $\frac{d}{dx}(x \ln(2))$

$$\begin{aligned}\frac{d}{dx}(x \ln(2)) &= \left(\frac{d}{dx} x \right)^{\nearrow 1} \cdot \ln(2) + x (\ln 2)^{\nearrow 0} \\ &= \ln(2)\end{aligned}$$

Example: $f(x) = 2^x$

Replace and simplify the result

$$\begin{aligned}\frac{d}{dx}2^x &= e^{x\ln(2)} \ln(2) \\ &= 2^x \ln(2)\end{aligned}$$

$$f'(x) = 2^x \ln(2)$$

Derivative of Exponential functions $f(x) = n^x$

Now, if we repeat the same process with $f(x) = n^x$ we get the following rule,

$$\frac{d}{dx} n^x = n^x \ln(n)$$

Class Activity

Compute the derivatives of the following functions,

$$\begin{aligned} f(x) &= \left(\frac{1}{2}\right)^x, & g(x) &= 3\left(\frac{1}{2}\right)^{x/2} \\ f'(x) &= -\frac{1}{2^x} \ln(2), & g'(x) &= -2^{-x/2-1} \ln(8) \end{aligned}$$

Power rule: $\frac{d}{dx} x^n = nx^{n-1}$

Chain rule: $\frac{d}{dx} u(v(x)) = u'(v(x)) \cdot v'(x)$

Quotient rule: $\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$

Product rule: $\frac{d}{dx} u(x)v(x) = u'v + uv'$

Simplifications using algebra

$$f(x) = \frac{x}{(x^2 + 1)^{3/2}}$$

$$f'(x) = \frac{1}{(x^2 + 1)^3} \left[(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1} \right]$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

Hint: Use a change of variable to simplify the expression to the eye.

Change of variable

Since $x^2 + 1$ repeats, let's introduce the following change of variable $a = x^2 + 1$.

$$f'(x) = \frac{1}{(x^2 + 1)^3} \left[(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1} \right] \rightarrow f'(x) = \frac{1}{a^3} \left[a^{3/2} - 3x^2 \sqrt{a} \right]$$

Use properties of exponents

Recalling the following properties of exponents (and notation): $\sqrt{x} = x^{1/2}$, $x^{n \cdot m} = (x^n)^m$ we can do the following,

$$\begin{aligned} f'(x) &= \frac{1}{a^3} \left[a^{3/2} - 3x^2 \sqrt{a} \right] \\ &= \frac{1}{a^3} \left[\left(a^{1/2} \right)^3 - 3x^2 a^{1/2} \right] \end{aligned}$$

Factor and use properties of exponents

Factor $a^{1/2}$.

$$\begin{aligned} f'(x) &= \frac{1}{a^3} \left[\left(a^{1/2} \right)^3 - 3x^2 a^{1/2} \right] \\ &= \frac{a^{1/2}}{a^3} \left[\left(a^{1/2} \right)^2 - 3x^2 \right] \end{aligned}$$

Use properties of exponents: $x^n = 1/x^{n-1}$, $x^n x^m = x^{n+m}$ and $(x^n)^m = x^{n \cdot m}$.

$$\begin{aligned} f'(x) &= \frac{1}{a^3 a^{-1/2}} \left[\left(a^{1/2} \right)^2 - 3x^2 \right] \\ &= \frac{1}{a^{3-1/2}} \left[a^{2/2} - 3x^2 \right] \end{aligned}$$

Finish the arithmetic and change notation

$$f'(x) = \frac{1}{a^{3-1/2}} \left[a^{2/2} - 3x^2 \right]$$

Since $3 - \frac{1}{2} = \frac{5}{2}$ and $\frac{2}{2} = 1$,

$$\begin{aligned} f'(x) &= \frac{1}{a^{5/2}} \left[a - 3x^2 \right] \\ &= \frac{1}{\sqrt{a^5}} \left[a - 3x^2 \right] \end{aligned}$$

Undo the change of variable

Now we can undo the change of variable, since we can't do anything else, $a = x^2 + 1$.

$$f'(x) = \frac{1}{\sqrt{a^5}} [a - 3x^2] \rightarrow f'(x) = \frac{1}{\sqrt{(x^2 + 1)^5}} [x^2 + 1 - 3x^2]$$

Final step: Simplify

$$\begin{aligned} f'(x) &= \frac{1}{\sqrt{(x^2 + 1)^5}} [1 - 2x^2] \\ &= \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}} \end{aligned}$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

Simplifications using algebra

Now, let's see how to reduce the following expression,

$$g(x) = 3 \left(\frac{1}{2} \right)^{x/2}$$

$$g'(x) = -3 \left(\frac{1}{2} \right)^{x/2} \ln \left[2^{1/2} \right]$$

$$g'(x) = -2^{-1-x/2} \ln [8]$$

Using properties of exponents

We start with the base of the exponential function, $\left(\frac{1}{2}\right)^{x/2}$

$$\begin{aligned}g'(x) &= -3 \left(\frac{1}{2}\right)^{x/2} \ln \left[2^{1/2}\right] \\&= -3 \frac{1^{x/2}}{2^{x/2}} \ln \left[2^{1/2}\right] \\&= -3 \frac{1}{2^{x/2}} \ln \left[2^{1/2}\right] \\&= -3 2^{-x/2} \ln \left[2^{1/2}\right]\end{aligned}$$

Using properties of logarithms

Let's move on with the logarithm, $a \ln [x] = \ln [x^a]$,

$$\begin{aligned} g'(x) &= -32^{-x/2} \ln [2^{1/2}] \\ &= -2^{-x/2} \ln \left[\left(2^{1/2} \right)^3 \right] \end{aligned}$$

Using properties of exponents

We are going to do an exponent exchange using the properties of exponents ($(a^n)^m = a^{n \cdot m} = (a^m)^n$). Recalling the expression, $g'(x) = -2^{-x/2} \ln \left[(2^{1/2})^3 \right]$ We are going to focus on $(2^{1/2})^3$.

$$(2^{1/2})^3 = 2^{3/2} = (2^3)^{1/2} = 8^{1/2}$$

Replace

Now that we know $(2^{1/2})^3 = 8^{1/2}$

$$\begin{aligned} g'(x) &= -2^{-x/2} \ln \left[(2^{1/2})^3 \right] \\ &= -2^{-x/2} \ln \left[8^{1/2} \right] \end{aligned}$$

The last step is to use properties of logarithms and exponents to get the expression $g'(x) = -2^{-1-x/2} \ln [8]$.

Properties of logarithms

Using the same property as before, $a \ln [x] = \ln [x^a]$, we can exchange the exponent of argument of the logarithm as a factor

$$\begin{aligned} g'(x) &= -2^{-x/2} \ln [8^{1/2}] \\ &= -2^{-x/2} \frac{1}{2} \ln [8] \end{aligned}$$

Properties of exponents

Using the following properties, $x^{-n} = 1/x^n$, $x = x^1$ and $x^n \cdot x^m = x^{n+m}$

$$\begin{aligned} g'(x) &= -2^{-x/2} \frac{1}{2} \ln[8] \\ &= -\frac{1}{2^{x/2}} \frac{1}{2} \ln[8] \\ &= -\frac{1}{2^{x/2+1}} \ln[8] \\ &= -2^{-x/2-1} \ln[8] \end{aligned}$$

End

$$f'(x) = \frac{(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1}}{(x^2 + 1)^3}$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

$$g'(x) = -3 \left(\frac{1}{2} \right)^{x/2} \ln \left[2^{1/2} \right]$$

$$g'(x) = -2^{-x/2-1} \ln [8]$$

Critical Points

What are the critical points?

Critical points are the roots of the derivative of a function or the values of the domain in which the derivative is undefined.

Why we are interested in critical points?

Critical points are essential for identifying potential local maxima, minima, or inflection points in the graph of the function.

Example: $f(x) = e^{\cos(x)}$

The derivative of the function $f(x) = e^{\cos(x)}$, by using the chain rule, is

$$f'(x) = -e^{\cos(x)} \sin(x).$$

Then to compute find the critical points we need to equate the derivative to zero and solve for x ,

$$f'(x) = 0 \leftrightarrow -e^{\cos(x)} \sin(x) = 0$$

Example: $f(x) = e^{\cos(x)}$

To solve $-e^{\cos(x)} \sin(x) = 0$ we have two possible solutions:

$$e^{\cos(x)} = 0, \quad \sin(x) = 0$$

$$\sin(x) = 0$$

$$e^{\cos(x)} = 0$$

The solutions of this equation are,

$$x = n\pi, \quad n \in \mathbb{Z}$$

$$\cos(x) = \ln(0)$$

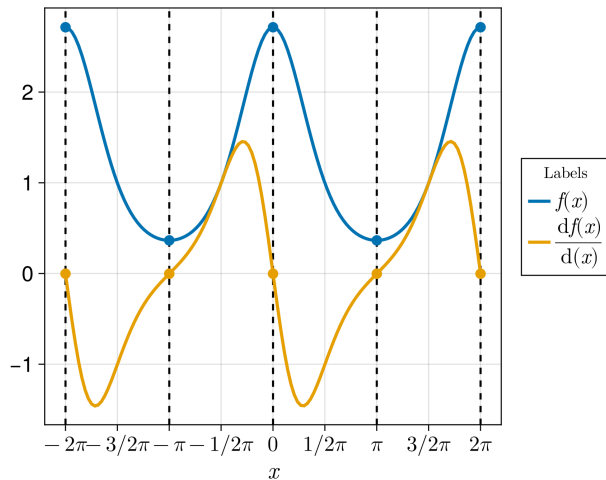
No solution.

Example: $f(x) = e^{\cos(x)}$

Since we have only solution for $\sin(x) = 0$ the critical points of the function $f(x)$ are,

$$x = n\pi, n \in \mathbb{Z}$$

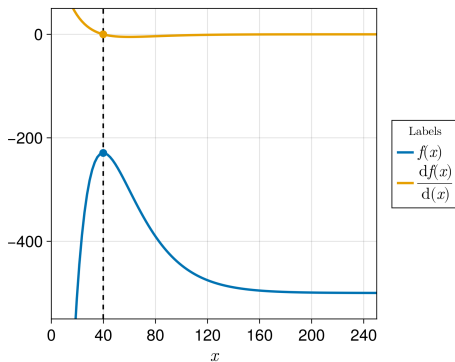
Example: $f(x) = e^{\cos(x)}$



Class activity

Compute the derivative of the following function. Then find its critical point.

$$P(x) = x100e^{-0.05x} - 20 \cdot 100e^{-0.05x} - 500$$



Critical Points: Class Activity

Compute the derivative of the functions

$$f(x) = xe^x$$

$$g(x) = -4.9x^2 + 98x + 10$$

Then find the critical points of the functions. Finally, evaluate the functions at the critical points in order to find the maximum or the minimum of the functions.

Derivative of logarithms.

Today's session we are going to learn how to compute the derivative of logarithms functions.

The overall methodology is as follows:

1. Use the change base formula
2. Compute the derivative of the change base formula
3. If possible, return to original base

Example: $f(x) = \log(x)$

Use the change base formula

$$\log(x) = \frac{\ln(x)}{\ln(10)}$$

Example: $f(x) = \log(x)$

Compute the derivative

$$\begin{aligned}\frac{d}{dx} \frac{\ln(x)}{\ln(10)} &= \frac{1}{\ln(10)} \frac{d}{dx} \ln(x) \\ &= \frac{1}{\ln(10)} \frac{1}{x} \\ &= \frac{1}{\ln(10)x}\end{aligned}$$

Example: $f(x) = \log(x)$

Return to original base

$$\begin{aligned}\frac{d}{dx} \frac{\ln(x)}{\ln(10)} &= \frac{1}{\ln(10)x} \\ &= \frac{1}{\frac{1}{\log(e)}x} \\ &= \frac{\log(e)}{x}\end{aligned}$$

Example: $f(x) = \log(x)$

Answers

$$f'(x) = \frac{1}{\ln(10)x} = \frac{\log(e)}{x}$$

Class Activity

Find the derivatives of the following logarithmic functions,

$$f(x) = \log_{\pi}(x)$$

$$g(x) = \log_2(2^x + 1)$$